

HDOC – 17

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PROBLEM STATEMENT:

Write a menu-driven C program using switch-case to perform digit-based operations on a positive integer. Define one separate user-defined function for each menu option, *with void return type* and *with a simple function call (without call by value/reference)*. Each function should read the required input, process it, and display the result. Prepare the algorithm and test case table. Then write, compile, execute, and test the C program against the prepared use cases.

- Generate the largest possible number using its digits
- Generate the smallest possible number using its digits.
- Swap its first and last digits.
- Check whether it is a palindrome number.

Algorithm:

1. Start the program.
2. Declare four user-defined functions:
 - largest() – generates the largest number using the digits.
 - smallest() – generates the smallest number using the digits.
 - swapDigits() – swaps the first and last digits.
 - palindrome() – checks whether the number is a palindrome.
3. Display the menu:
 - 1 → Largest possible number
 - 2 → Smallest possible number

- 3 → Swap first and last digits
 - 4 → Check palindrome
4. Read the user's choice.
 5. Use switch(choice):
 - If choice is 1, call largest().
 - If choice is 2, call smallest().
 - If choice is 3, call swapDigits().
 - If choice is 4, call palindrome().
 - Otherwise, display "Invalid choice".
 6. Each function takes the positive integer from the user, performs its operation, and displays the result.
 7. Stop.

Test Case table:

Test case	Choice	Input	Expected output	Actual output	Result
1.	1	12054	Largest number = 54210	Largest number = 54210	Pass
2.	2	4021	Smallest number = 1024	Smallest number = 1024	pass
3.	3	9876	Number after swapping = 6879	Number after swapping = 6879	pass
4.	4	12321	12321 is a palindrome number	12321 is a palindrome number	pass
5.	5	_____	Invalid choice	Invalid choice	pass

C Program:

The image shows a screenshot of a C program being edited in a virtual machine environment. The window title is 'kali-linux-2026.2-virtualbox-amd64 [Running] - Oracle VirtualBox'. The menu bar includes 'File', 'Machine', 'View', 'Input', 'Devices', and 'Help'. Below the menu is a toolbar with icons for file operations and a tab bar showing tabs 1 through 4. The main editing area has a dark background with light-colored text. The code defines two functions, 'largest()' and 'smallest()', both of which take an integer 'n' and a frequency array 'freq[10]'. Both functions use a while loop to process the digits of 'n' and a for loop to print the frequency of each digit. The 'largest()' function prints the largest number by iterating from 9 down to 0, while the 'smallest()' function prints the smallest number by iterating from 0 up to 9. The code is as follows:

```
43     return 0;
44 }
45
46
47 void largest()
48 {
49     int n, digit, freq[10] = {0};
50
51     printf("Enter a positive integer: ");
52     scanf("%d", &n);
53
54     while(n > 0)
55     {
56         digit = n % 10;
57         freq[digit]++;
58         n = n / 10;
59     }
60
61     printf("Largest number = ");
62
63     for(int i = 9; i >= 0; i--)
64     {
65         while(freq[i] > 0)
66         {
67             printf("%d", i);
68             freq[i]--;
69         }
70     }
71
72     printf("\n");
73 }
74
75
76 void smallest()
77 {
78     int n, digit, freq[10] = {0};
79
80     printf("Enter a positive integer: ");
81     scanf("%d", &n);
82
83     while(n > 0)
84     {
85         digit = n % 10;
86         freq[digit]++;
87         n = n / 10;
88     }
```

```
kali-linux-2026.2-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

1 2 3 4

Open

82
83 while(n > 0)
84 {
85     digit = n % 10;
86     freq[digit]++;
87     n = n / 10;
88 }
89
90 printf("Smallest number = ");
91
92 for(int i = 1; i <= 9; i++)
93 {
94     if(freq[i] > 0)
95     {
96         printf("%d", i);
97         freq[i]--;
98         break;
99     }
100 }
101
102 for(int i = 0; i <= 9; i++)
103 {
104     while(freq[i] > 0)
105     {
106         printf("%d", i);
107         freq[i]--;
108     }
109 }
110
111 printf("\n");
112 }
113
114
115 void swapDigits()
116 {
117     int n, first, last, temp, power = 1, result;
118
119     printf("Enter a positive integer: ");
120     scanf("%d", &n);
121
122     last = n % 10;
123     temp = n;
124
125     while(temp >= 10)
126     {
```

```
kali-linux-2026.2-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
1 2 3 4
Open
115 void swapDigits()
116 {
117     int n, first, last, temp, power = 1, result;
118
119     printf("Enter a positive integer: ");
120     scanf("%d", &n);
121
122     last = n % 10;
123     temp = n;
124
125     while(temp >= 10)
126     {
127         temp = temp / 10;
128         power = power * 10;
129     }
130
131     first = temp;
132
133     result = n - first * power - last;
134     result = result + last * power + first;
135
136     printf("Number after swapping = %d\n", result);
137 }
138
139
140 void palindrome()
141 {
142     int n, temp, digit, reverse = 0;
143
144     printf("Enter a positive integer: ");
145     scanf("%d", &n);
146
147     temp = n;
148
149     while(temp > 0)
150     {
151         digit = temp % 10;
152         reverse = reverse * 10 + digit;
153         temp = temp / 10;
154     }
155
156     if(n == reverse)
157         printf("%d is a palindrome number.\n", n);
158     else
159         printf("%d is not a palindrome number.\n", n);
160 }
```

```
kali-linux-2026.2-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

1 #include <stdio.h>
2
3 void largest();
4 void smallest();
5 void swapDigits();
6 void palindrome();
7
8 int main()
9 {
10     int choice;
11
12     printf("----- MENU ----- \n");
13     printf("1. Generate largest number \n");
14     printf("2. Generate smallest number \n");
15     printf("3. Swap first and last digits \n");
16     printf("4. Check palindrome \n");
17
18     printf("Enter your choice: ");
19     scanf("%d", &choice);
20
21     switch(choice)
22     {
23         case 1:
24             largest();
25             break;
26
27         case 2:
28             smallest();
29             break;
30
31         case 3:
32             swapDigits();
33             break;
34
35         case 4:
36             palindrome();
37             break;
38
39         default:
40             printf("Invalid choice \n");
41     }
42
43     return 0;
44 }
45
46
```

Compile the program using:

```
gcc hdoc17.c -o hdoc17
```

Execute the code using:

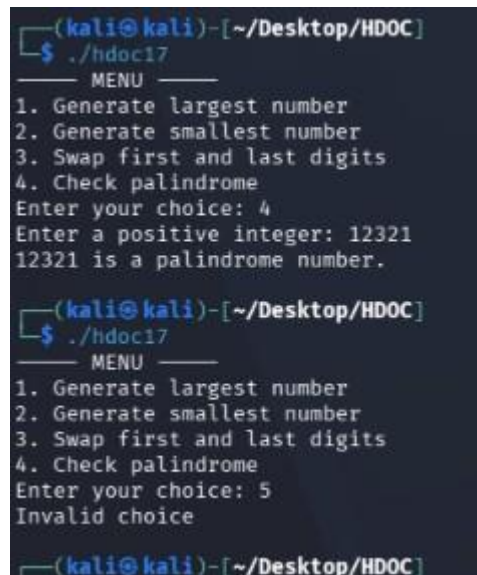
```
./hdoc17
```

Here are some screenshots of compilation and execution against test case:



```
kali-linux-2026.2-virtualbox-amd64 [Running] - Ora
File Machine View Input Dev

Session Actions Edit View Help
(kali@kali)-[~/Desktop]
$ cd HDQC
(kali@kali)-[~/Desktop/HDQC]
$ gedit hdoc17.c
** (gedit:2205): WARNING **: 12:23:16.298: Co
n '1.0' not found
** (gedit:2205): WARNING **: 12:23:16.298: Co
version '1.0' not found
(kali@kali)-[~/Desktop/HDQC]
$ gcc hdoc17.c -o hdoc17
(kali@kali)-[~/Desktop/HDQC]
$ ./hdoc17
MENU
1. Generate largest number
2. Generate smallest number
3. Swap first and last digits
4. Check palindrome
Enter your choice: 1
Enter a positive integer: 12054
Largest number = 54210
(kali@kali)-[~/Desktop/HDQC]
$ ./hdoc17
MENU
1. Generate largest number
2. Generate smallest number
3. Swap first and last digits
4. Check palindrome
Enter your choice: 2
Enter a positive integer: 4021
Smallest number = 1024
(kali@kali)-[~/Desktop/HDQC]
$ ./hdoc17
MENU
1. Generate largest number
2. Generate smallest number
3. Swap first and last digits
4. Check palindrome
Enter your choice: 3
Enter a positive integer: 9876
Number after swapping = 6879
```



```
(kali@kali)-[~/Desktop/HDQC]
$ ./hdoc17
MENU
1. Generate largest number
2. Generate smallest number
3. Swap first and last digits
4. Check palindrome
Enter your choice: 4
Enter a positive integer: 12321
12321 is a palindrome number.
(kali@kali)-[~/Desktop/HDQC]
$ ./hdoc17
MENU
1. Generate largest number
2. Generate smallest number
3. Swap first and last digits
4. Check palindrome
Enter your choice: 5
Invalid choice
(kali@kali)-[~/Desktop/HDQC]
```